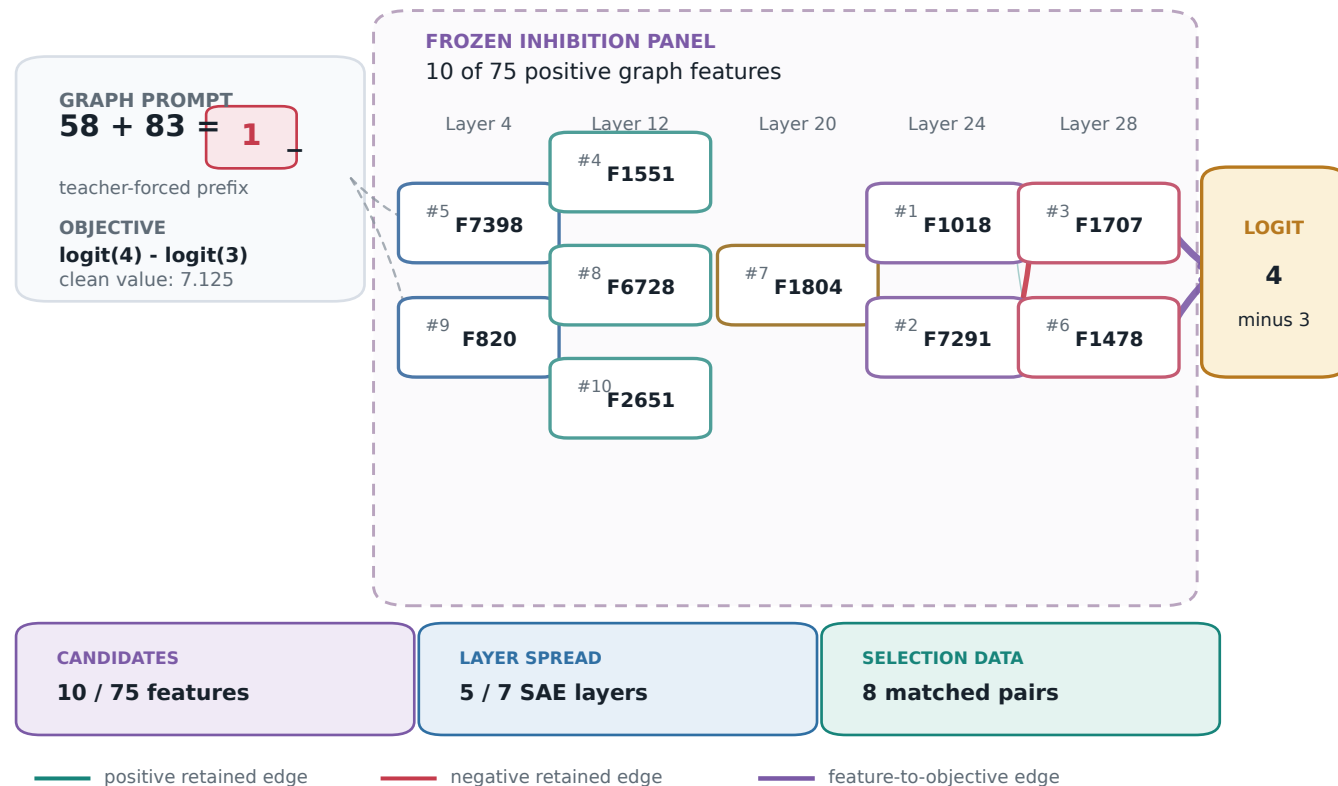


# Arithmetic feature inhibition affects digit state, not carry specifically

Attribution-guided TopK SAE panel and frozen matched-control confirmation

## A Candidate panel from one carry graph



## B Frozen causal specificity test

### INTERVENTION

**INHIBIT THE SAME 10 LATENTS**

$z_j < 0$  at the final token; reconstruction error preserved

**CARRY TARGETS**

24 held-out cases

**NO-CARRY CONTROLS**

24 matched controls

MEAN CHANGE IN CORRECT-MINUS-CONTRAST GAP

carry -0.141

control -0.109

**FROZEN TOP-10 CONFIRMATION**

**Paired specificity**

**-0.031**

Bootstrap 95% CI

**[-0.089, +0.026]**

21/24 carry effects negative

10/24 beat control

Token flips

**0 carry | 0 control**

**PREDECLARED CARRY-SPECIFICITY RULE: NOT MET**

Interpretation: inhibition produced a reproducible directional arithmetic effect, but the matched no-carry effect was nearly as large.

The panel is therefore carry-associated at discovery, not an identified abstract carry variable. Solid arrows are retained graph edges; dotted input paths compress displayed token edges.